CONCEPT NOTE

ResumeMatch AI
Career Guidance Agent

SDG 8: Decent Work and Economic Growth

IBM SkillsBuild AI/ML Internship

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1. Introduction & Context

India produces a large number of engineering and professional graduates every year. Yet many skilled individuals living in Tier-2 and Tier-3 cities struggle to convert their education into decent employment. Common barriers include poorly tailored resumes that fail applicant-tracking systems, limited access to professional career counselling, and uncertainty about which skills to acquire next. This project addresses these gaps with an offline, free AI-powered career guidance agent aligned to **Sustainable Development Goal 8 – Decent Work and Economic Growth**.

2. Problem Statement

"Many skilled individuals in Tier-2/3 cities face employment challenges. This AI project will suggest personalized job paths based on user skills and interests."

More specifically, three interlocking problems are targeted:

- Resumes are frequently rejected because they miss the exact keywords and skills present in the job description.
- Job seekers in smaller cities have little affordable access to quality career counselling.
- Students and fresh graduates often do not know which skills to learn next to become job-ready.

3. Objectives

- Build an offline tool that computes a transparent Resume–Job Description match score.
- Highlight missing skills drawn from a curated Indian job-market skill bank and give three actionable improvement tips.
- Recommend a suitable career path and a personalised learning plan.
- Keep the solution free, lightweight and easy for colleges and placement cells to adopt.

4. Proposed Solution – ResumeMatch AI

ResumeMatch AI is a pure-Python application that runs entirely on the user's machine. It combines classic information-retrieval techniques with a domain-specific skill ontology:

4.1 Match Score – Hybrid score = 55% TF-IDF cosine similarity + 45% skill-overlap (Jaccard). The result is presented as a clear percentage.

4.2 Missing Skills – Skills present in the job description but absent from the resume are listed, prioritised from a bank covering programming, web, data/ML, cloud/DevOps, soft skills and Indian domain keywords.

4.3 Improvement Tips – Three concrete, prioritised suggestions generated from the score band and the missing-skill list.

4.4 Career Path & Learning Plan – Based on detected skills the system maps the user to one of five high-demand paths (Software Engineer, Data Analyst, ML Engineer, Frontend Developer, DevOps) and supplies a four-step personal learning plan.

High-level concept: *“Grammarly for resumes, with a career coach built in.”*

5. Lean Canvas

The business and product logic is summarised in the Lean Canvas below (identical in content to the submitted canvas). A full-page landscape Lean Canvas is also provided as a separate PDF.

PROBLEM
• Resumes get rejected because they miss the keywords and skills in the job description.
• Job seekers in Tier-2/3 cities have little access to career counselling.
• Students do not know which skills to learn next.

EXISTING ALTERNATIVES
• Paid resume-checker websites
• Generic chatbots
• Advice from seniors or YouTube

SOLUTION
• Resume vs job description match score (TF-IDF)
• Missing-skills list and 3 improvement tips
• Career-path matching and a personal learning plan

KEY METRICS
• Resumes analysed
• Average score gain after edits
• Repeat users
• Interview calls reported

UNIQUE VALUE PROPOSITION
• A free tool that tells you in seconds how well your resume fits a job, and exactly what to fix and learn next.

HIGH-LEVEL CONCEPT
• Grammarly for resumes, with a career coach built in

UNFAIR ADVANTAGE
• Skill list built around the Indian job market
• Runs offline, no API key or cost
• Easy for colleges to adopt

CUSTOMER SEGMENTS
• Final-year students and fresh graduates
• Skilled job seekers in Tier-2/3 cities

EARLY ADOPTERS
• Final-year engineering students applying for first internships and jobs

CHANNELS
• College placement cells and clubs
• LinkedIn and WhatsApp groups
• GitHub repository

COST STRUCTURE
• Hosting and domain
• Development and maintenance time
• Optional AI API costs for future features
• Minimal marketing (campus word of mouth)

REVENUE STREAMS
• Free core tool for students
• Premium features (deeper analysis, saved history)
• Licences for colleges, placement cells and training institutes

6. Technical Approach

The prototype is implemented in Python 3 with scikit-learn for TF-IDF vectorisation and cosine similarity. No external API keys or internet connection are required at runtime. Key design choices:

- **Offline-first** – all computation happens locally; privacy is preserved and cost is zero.
- **Transparent scoring** – users can understand why a score is high or low (keyword + skill contribution).
- **Indian-market skill bank** – curated list of languages, frameworks and soft skills commonly requested by Indian employers.
- **Extensible** – future versions can add PDF parsing, a lightweight web UI (Streamlit/Flask), or optional embedding models.

Input: plain-text resume + plain-text job description.
Output: console report, optional HTML report and JSON result containing match percentage, skill lists, tips and learning plan.

7. Target Users & Expected Impact

Primary users are final-year students and fresh graduates, especially those studying or living in Tier-2 and Tier-3 cities. Secondary users include college placement cells and training institutes that can deploy the tool at scale.

By giving every student an immediate, actionable diagnosis of their resume and a clear learning roadmap, the project contributes directly to SDG 8 targets of higher youth employment and reduced inequality of opportunity.

8. Implementation Status (Prototype)

- Working CLI application with hybrid TF-IDF + skill-overlap scoring.
- Skill extraction, missing-skill detection, three tips and five career-path learning plans.
- HTML report generator and JSON export.
- Sample resume and job description for demonstration.
- Fully documented repository ready for college adoption.

9. Future Roadmap

- PDF and DOCX resume parsing.
- Simple web interface (Streamlit or Flask) for non-technical users.
- Optional deeper semantic matching with sentence-transformers (still offline).
- Multi-language support (English + Hindi).
- College licence dashboard and anonymised analytics for placement cells.

10. Conclusion

ResumeMatch AI demonstrates that a focused, offline AI tool can meaningfully improve the job-search experience of students who currently lack access to expensive counselling or paid resume services. The prototype is functional, transparent and aligned with both the technical goals of the IBM SkillsBuild AI/ML Internship and the broader societal goal of decent work for all (SDG 8).

Lean Canvas: ResumeMatch AI Career Guidance Agent

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